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COIMBATORE CAMPUS

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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-AOH-I105*

Team Name: *Dark Protocol*

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Presentation Outline

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Problem Statement

To develop a **cooperative-owned digital service marketplace platform** that enables Labour Cooperative Federations and Labour Cooperative Societies to provide **verified household and community services** while ensuring **fair wages, worker welfare and consumer trust**.

Labour Cooperative Federations and Societies have a large pool of skilled workers (electricians, plumbers, carpenters, painters, domestic helpers, caregivers, drivers, gardeners, cleaners and technicians), but **no structured digital platform** to connect them with households and institutions that need these services.

Problem Understanding

- **Who is affected:** cooperative workers and households or institutions seeking services. Over 90% of India's workforce is in the unorganised sector and invisible online.
- **Cause:** private platforms dominate the market, while cooperative workers remain underutilised despite their skills and local presence. Federations vouch for worker skill and conduct, but this trust has no digital form.
- **Effect on workers:** platform commissions of up to 30% with no insurance, and platform work sits outside ESIC/EPFO protection.
- **Closing window:** India's gig workforce grows from 7.7M (2020–21) to 23.5M by 2029–30. Cooperatives risk losing members to commission-heavy platforms.

Existing Gaps & Challenges

- **Commission extraction:** Urban Company charged workers up to 30% commission; protests reached Delhi, Bengaluru and Mumbai as recently as Feb 2026.
- **No safety net:** Urban Company alone runs 52,000+ workers, a third of them women, with no guaranteed insurance.
- **No digital trust:** no verification, profiling or accountability that a customer can see, even though federations already vouch for their workers.
- **Digital exclusion:** many cooperative workers do not own a smartphone or are not comfortable using apps.
- **Unsolved challenge:** no cooperative-owned platform combines verified workers, fair wages and welfare for household and community services.

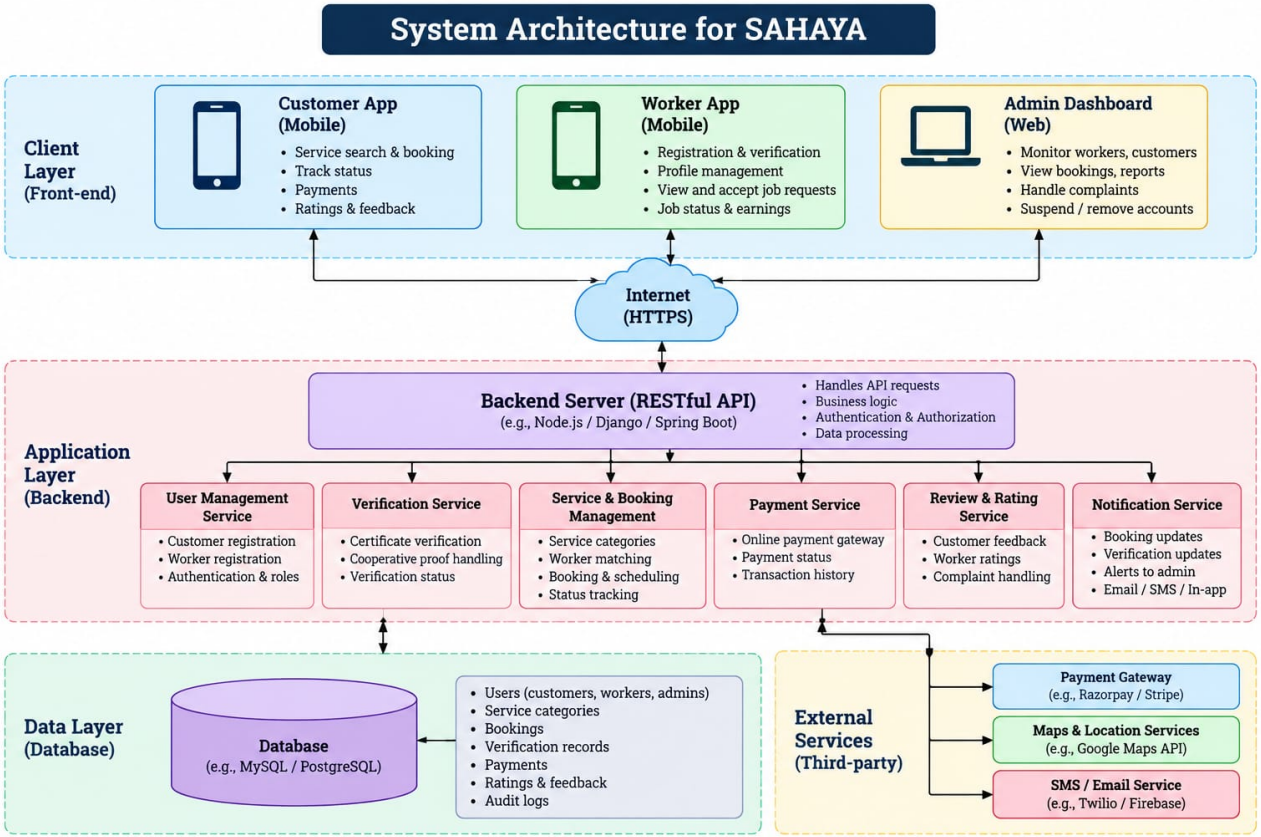
Our Proposed Solution

SAHAYA: a cooperative-owned digital marketplace with three connected parts and a toll-free IVR.

- **Customer App:** register, browse a service, pick date and time, compare verified workers (profile, rating, availability, price), book, track status, then pay online or by cash and rate the worker.
- **Worker App:** registration with verification through a certificate, or through a cooperative role-assurance proof for workers without one; then profile setup, job requests, earnings and booking history.
- **Admin Dashboard:** the federation monitors workers, customers, bookings and complaints, views reports, and acts on emergencies or policy violations with an audit log.
- **Toll-Free IVR:** workers without app access begin registration by phone call.

Only verified workers are shown to customers; the cooperative retains ownership of the platform and its data.

Technical Approach



Tech Stack

Frontend: Flutter (Customer and Worker apps), React (Admin)

Backend: Node.js / Django REST APIs, Supabase Auth + JWT + RBAC

Database: PostgreSQL + PostGIS (Supabase)

AI/ML, APIs, or Hardware: Python (Prophet / LSTM, Scikit-learn); Razorpay / UPI; Asterisk / Twilio (IVR + SMS); AWS / GCP

Innovation & Uniqueness

- **IVR + App reach:** the only cooperative platform pairing an app with a toll-free IVR, reaching workers with no smartphone or data.
- **Zero commission:** cooperative-owned, not privately held.
- **Sahaya Shield:** booking levy and donations fund a cooperative-owned welfare pool, with no insurer middleman.
- **Federation-verified trust:** workers are verified against real cooperative records, not self-reported profiles.
- **AI forecasting for workers:** predicts demand and pre-allocates work by zone, aimed at worker income rather than platform margin.

Research backing: **(1)** Bunders et al., 2022: explains the "capital conundrum" and why local odd-job co-ops succeed better than global ones. **(2)** Kataru, 2026: links digital co-op architectures to India's 2020 Labour Codes. **(3)** Socio-Economic Review Cluster, 2026: co-op platforms eliminate predatory algorithmic management.

Feasibility & Scalability

- **Feasibility:** built on proven, mature technology; existing federation records give instant worker verification with no cold-start trust problem; federations already have ground presence and member trust.
- **Scalability:** cloud-native microservices scale on demand at low upfront cost; phased, zone-wise rollout tunes the AI models before scaling.

Risks and mitigations:

Risk	Mitigation
Low smartphone access and digital literacy	Toll-free IVR removes the barrier
Building first-time trust among offline-first workers	Federation-led onboarding drives by trusted local field staff
Verification standards across independent federations	Standardised SOPs and a central admin dashboard
Geo-matching and AI forecasting accuracy across regions	Phased, zone-wise rollout using local demand data
Privacy and security of worker and consumer data	Role-based access, encryption, secure S3 storage
Sustaining welfare-pool funding	Levy, donations and CSR/government partnerships

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate] (cloud-native microservices keep upfront infrastructure cost low)
- Revenue model: booking levy and donations sustain the Sahaya Shield welfare pool without high commissions
- Estimated total: [Add amount and currency]

Use Cases (compulsorily 3 use cases must be there)

- **Use case 1, household customer:** books a service, sees only verified workers, compares rating, availability and price, tracks status (Accepted, En Route, In Progress), then pays online or by cash and rates the worker.
- **Use case 2, worker without a certificate:** contacts the Labour Cooperative, obtains a role-assurance proof, uploads it for validation, then sets up a profile and receives job requests.
- **Use case 3, worker without a smartphone:** calls the toll-free IVR, chooses language and service, enters a pincode, visits the nearby Panchayat within one week, and is registered on e-Shram if not part of any Federation.

Impacts and Benefits

Impacts: what changes?

- **Digital visibility:** a verifiable digital identity and service footprint.
- **Better utilisation:** job allocation connects idle skills to local demand.
- **Fairer opportunities:** transparent, workload-aware distribution of jobs.
- **Data-driven co-ops:** insights on demand, performance and skill gaps.

Benefits: who benefits?

- **Workers:** more work, verifiable history, fair earnings, welfare support.
- **Cooperatives:** streamlined operations, demand insights, levy revenue for welfare.
- **Consumers:** verified workers, transparent pricing and tracking, emergency services.
- **Communities:** stronger local service ecosystem and greater inclusion.

Ultimate outcome: a transparent, inclusive and data-driven cooperative service ecosystem that creates value for all.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 8	Target 8.5	Decent Work and Economic Growth	Fair wages and more work opportunities for cooperative workers
SDG 1	Target 1.3	No Poverty	Sahaya Shield welfare pool gives workers a safety net
SDG 10	Target 10.2	Reduced Inequalities	Toll-free IVR includes workers without smartphones

Individual Member Contribution

Member	Contribution or responsibility
Nithin	Customer App
M Pavna Preethikha	Backend Development
Sattineni Venkata Satya Tulasi Ram	Worker App
Nehasri Mullapudi	Admin Dashboard and Toll-Free IVR

Yukti ID

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